

INTRODUCTION TO STOCHASTIC OPTIMIZATION

Random Search, Simulated Annealing, Evolutionary Strategies, Genetic Algorithms

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Stochastic Optimization

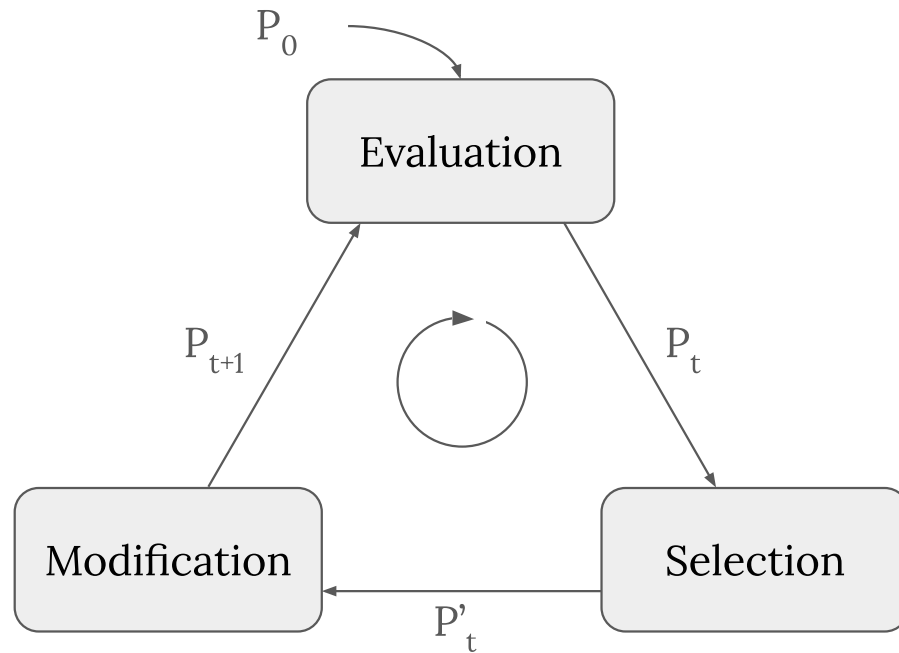
<https://supaerodatascience.github.io/stochastic/>



STOCHASTIC OPTIMIZATION

- Search methods which use random variables to optimize on an objective function
- No requirements for the objective function: not differentiable, convex, concave, simple. Any function works.
- This is not problems which are stochastic (stochastic programming)
- Other names: metaheuristics, black-box optimization, stochastic search
- Examples: simulated annealing, particle swarm optimization, evolutionary strategies, genetic algorithms

EVOLUTIONARY ALGORITHMS



Population of solutions are evaluated, selected, then modified to create the next population.

WHY DO THEY WORK?

- They work by testing several solutions, until they find a solution that works.
- They don't require domain knowledge or specific characteristic of the problems.
- These methods are based on random process.
 - Randomness can also make the method less sensitive to modeling errors.
 - Randomness may enable the method to escape a local optimum.

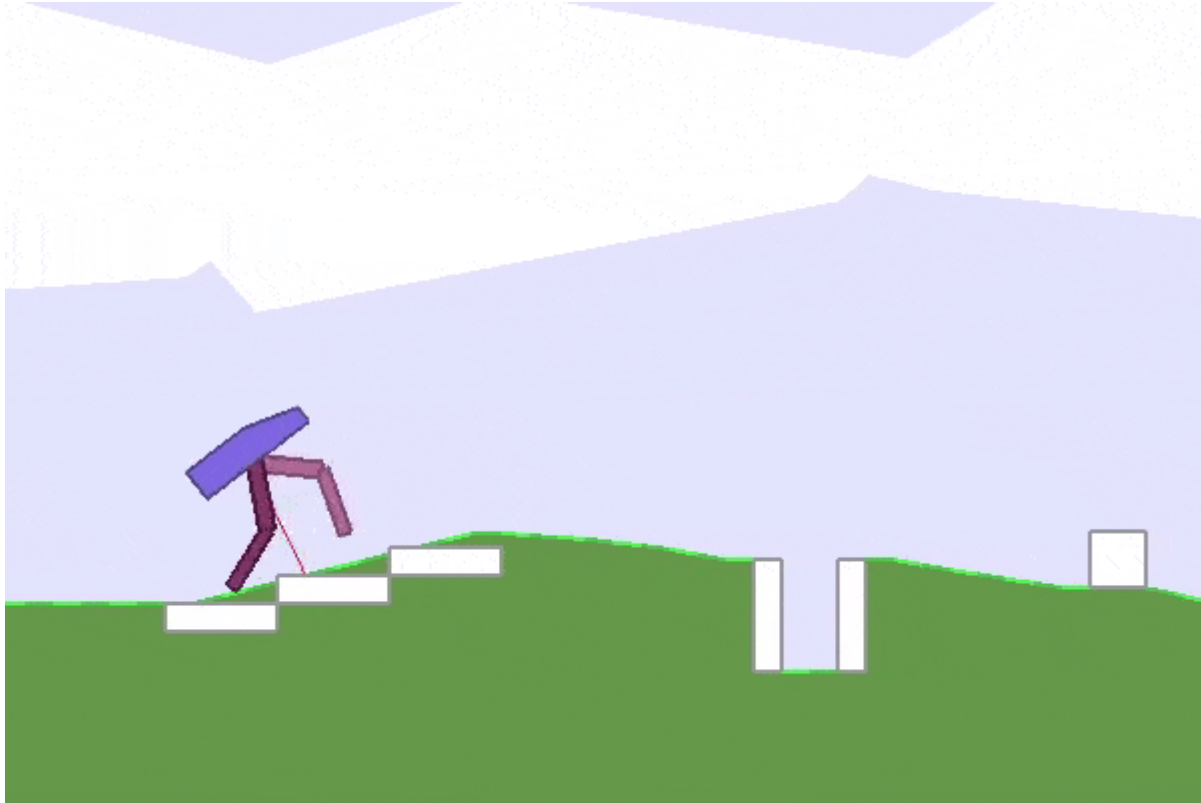
WHY ARE THEY GREAT?



"This evolved antenna represents the world's first artificially-evolved object to fly in space."

https://en.wikipedia.org/wiki/Evolved_antenna

WHY ARE THEY GREAT?

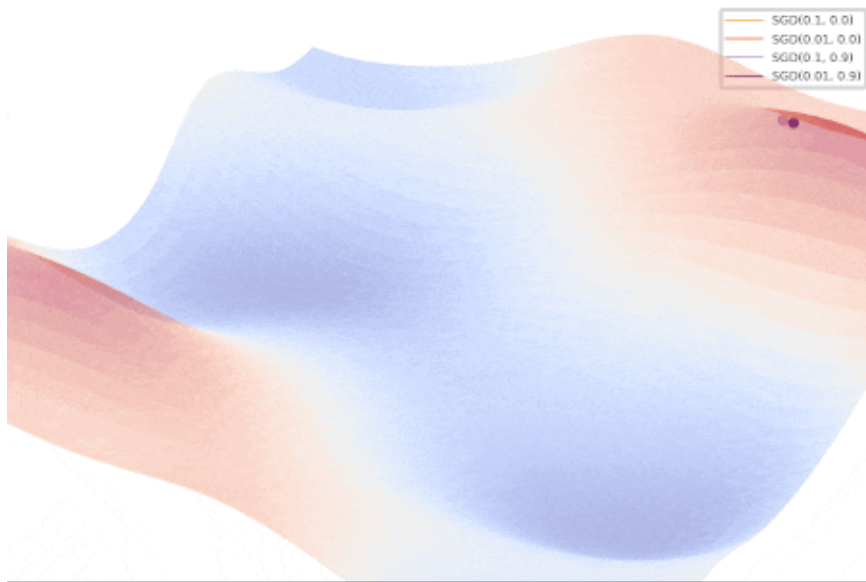


Robotic control with evolved logic

Representation: list of cities in order

Objective function: total distance traveled

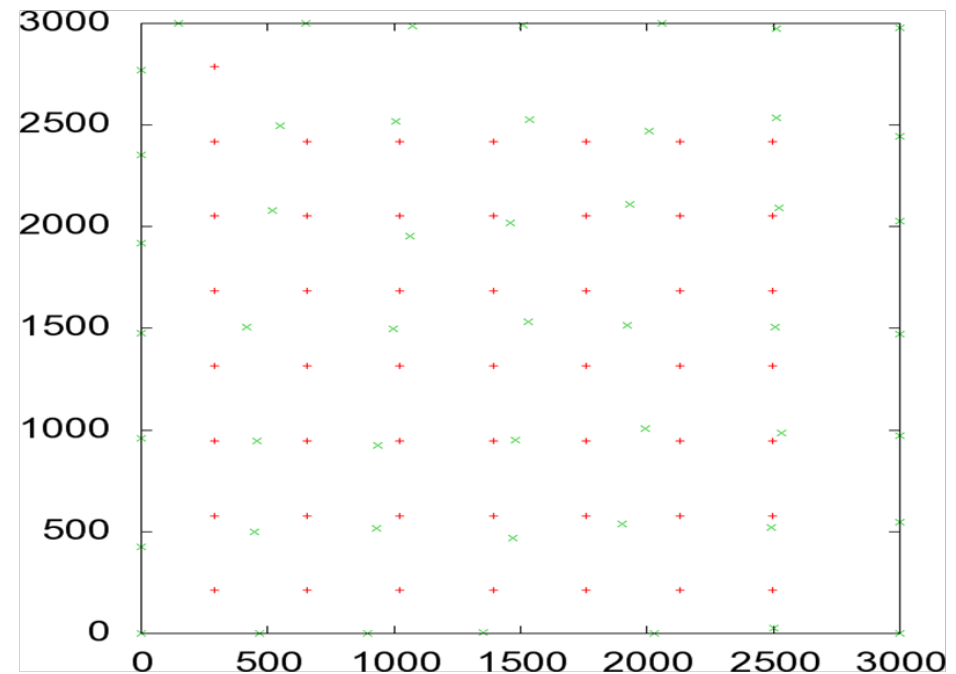
SEARCH: CONTINUOUS OPTIMIZATION



Representation: coordinates in N dimensional space

Objective function: problem-specific

APPLICATION: WIND FARM LAYOUT OPTIMIZATION



Representation: x,y coordinates of wind turbines

Objective function: Expected energy gain

Stochastic search methods are state-of-the-art because resolution methods based on constraint satisfaction or planning are too costly given the physical simulation of wind turbine wakes.

[illegible]

Objective function: Runtime, number of tests passed

ADVANTAGES

- Creative
- Easy to implement and parallelize
- Flexibility in problem representation and in fitness function
- Many proven algorithms with (comparatively) few hyperparameters
- Relatively easy to understand

DISADVANTAGES

- Computational cost
- Search not fully informed by fitness function
- Not guaranteed to find the optimal individual

CLASS FORMAT

- Course website at <https://SupaeroDataScience.github.io/stochastic/>
- Notebooks available on course website or at <https://github.com/SupaeroDataScience/stochastic/>
- Evaluation by quizzes at the beginning of second and third class

CLASS SCHEDULE

Schedule

03/03	<u>Random search</u>	Continuous optimization, random search, simulated annealing
05/03	<u>Evolutionary Strategies</u>	Population-based methods, 1+1 ES, CMA-ES
05/03	<u>Genetic Algorithms</u>	Genetic Algorithm, Multi-Objective Optimization, NSGA-II

EXERCISE: RECENT APPLICATIONS

1. **Heuristic Strategies for Solving Complex Interacting Stockpile Blending Problem with Chance Constraints**
2. **Design of Specific Primer Sets for SARS-CoV-2 Variants Using Evolutionary Algorithms**
3. **Continuously Running Genetic Algorithm for Real-Time Networking Device Optimization**
4. **Evolutionary Minimization of Traffic Congestion**
5. **From Pixels to Metal: AI-Empowered Numismatic Art**
6. **Evolutionary design of explainable algorithms for biomedical image segmentation**
7. **Freeform generative design of complex functional structures**
8. **Learning Traffic Signal Control via Genetic Programming**

What is the application in the article? How is this application formulated into an evolutionary fitness?

What is the method used in the article? Describe it in simple terms.